Final Project

Assignment Description

Due 11:59 PM on April 29

1 Overview of Assignment

For your final project, you will work on a program that demonstrates an advanced physical simulation topic of your choice. Some suggestions are listed below, but you are also free to propose your own topic. You will implement the project, present a demo, and prepare a brief writeup describing the work you have done.

1.1 Reminder about Teams

Per the syllabus, you may work on this project in self-selected teams of up to two. If you work in a team, you must each submit a collaboration report, as in previous assignments.

1.2 Starter Code

Since the final project is open-ended, there is no starter code. You may use any code you have written in previous projects as a starting point for the final project, and you may also use external libraries or codebases, if approved by the instructor in advance.

2 Submission Instruction

The final project has multiple components. The presentations will take place during the class final exam period; your attendance and participation at the final exam presentation is mandatory. All other project materials (including the writeup, code, and any other artifacts such as videos or screen captures) must be submitted to canvas **before** the start of the final project presentations. Include a README in your submission explaining its contents. Late days may not be used on the final project; materials submitted after the final project presentations will not be graded.

3 Grading

The final project grade will be determined by your score on three components: the implementation itself, the presentation, and a written report.

4 Project Implementation *(100 points)*

Decide on a project to work on, and provide an implementation of the project. The project should highlight an advanced topic relating to physical simulation that was *not* covered in the previous assignments, and whose depth and complexity exceeds the previous assignments. The following list of topics are "pre-approved" starting points; if you want to work on one of these topics, you may do so without taking any additional steps. You can also pick your own topics to work on. If you do, you are **strongly encouraged** to discuss your plans for the final project with me before you begin working, to make sure the scope and topic is appropriate and likely to receive full credit.

- Extend the cloth simulation project, adding handling of collisions between the cloth and itself, and two-way interactions between cloth and rigid bodies. A good starting point is *Robust Treatment of Collisions, Contact and Friction for Cloth Animation* by Bridson et al.
- Implement a fluid simulation. There are two fundamentally different approaches to fluid simulation, one involving a fixed grid (*Eulerian fluids*) and one involving moving particles of fluid (*smoothed particle hydrodynamics*). Both approaches are acceptable but the former is probably the most accessible. As a starting point, see *Realistic Animation of Liquids* by Foster and Metaxas.
- Research how to simulate stable stacking of rigid bodies with friction, and implement a demo. The work of David Baraff may prove useful.
- Implement a real-time simulation of squishy bodies using dimensional model reduction to boost performance. See *Real-Time Subspace Integration for St. Venant-Kirchhoff Deformable Models* by Barbič and James for inspiration.
- Implement a hair simulation, taking into account the stretching, bending, and twisting resistance of the hair. See for instance *Discrete Elastic Rods* by Bergou et al or *Super-Helices for Predicting the Dynamics of Natural Hair* by Bertails et al.

You must submit full source code for your project. In addition, you must submit gradeable artifacts that demonstrate the project's key features; these artifacts might include

- exectuable demos that the instructor can run, with *complete* instructions included in a README for how to compile, run, and operate the demos. Any executable demos must compile and run on all three major operating systems (Ubuntu Linux, OS X, and Windows) without specialized hardware beyond a modern graphics card;
- screen captures of you operating an executable demo, as you talk through your program's major features;
- pictures, videos, or 3D models generated using your program.

You are strongly encouraged to discuss with the instructor, well in advanace, your plan for which research artifacts you will submit, to ensure that they will be sufficient to allow grading of your final project.

5 Project Demo *(50 points)*

You will have a chance to present a demo of your finished project to the instructor and the class, during the final exam period. Details of the demo and its logistics will be announced closer to the project deadline.

6 Project Writeup *(50 points)*

Please prepare a report with details about the project you worked on. The report should be targeted at an audience that understands the basics of physical simulation (i.e., has sat in on the class lectures) but does not know the details of the specific algorithms and techniques you implemented. Briefly describe the goals of the project you worked on, and how the key algorithms and component work. Provide any mathematical derivations or equations needed to understand how your project's solution works.

In a second section, briefly describe how your implementation is engineered: give enough hints that the instructor and TA can figure out how your code works. Also describe any known bugs/limitations with your implementation.

The report should be submitted along with your code; it must be a PDF or Word file and each section should be **at most two pages long** (so the total report will be a maximum of four pages long). You may include additional figures or videos (see the section on project artifacts above) in your submission and these will not count against your report page limit.

7 Extra Credit *(up to 30 points)*

Projects that are particularly impressive in scope and difficulty will earn up to 30 points of extra credit. You don't need to take any extra steps to be eligible for this extra credit—other than showing off your work in the best possible light in your presentation/demo and writeup.

8 Extra Credit *(10 points)*

The university allows me to give 10 points of extra credit to any team member who completes the online course instructor survey. Include a statement in the README affirming, on your honor, that you have submitted the survey.

9 Reminder about Collaboration Policy and Academic Honesty

You are free to talk to other students about the algorithms and theory involved in this assignment, or to get help debugging, but all code you submit (except for the starter code, and the external libraries included in the starter code) must be your own. Please see the syllabus for the complete collaboration policy.